

RITESH KUMAR

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[LinkedIn](#) | [GitHub](#) | [LeetCode](#) | [Portfolio](#)

TECHNICAL SKILLS

Languages: Java, JavaScript (Node.js), Python, C++, SQL
Backend: Node.js, Express.js, REST APIs, JWT & OAuth, WebSockets, Microservices, Event-Driven Architecture
Databases: MongoDB (Atlas, Aggregation, Indexing), MySQL, Redis (caching)
Cloud & Tools: AWS (EC2, S3, IAM — basics), Git, GitHub, Linux
Tools & APIs: FFmpeg, OpenAI Whisper, OpenRouter, Cloudinary, Postman, Chart.js
CS Foundations: DSA, OOP, DBMS, Operating Systems, Computer Networks, System Design (basics)

EXPERIENCE

Backend Developer Intern — Mentor Connect (Govt. of Punjab)

Jun 2025 – Jul 2025

Government of Punjab — Remote

- Re-architected MongoDB data models for the Mentor Connect platform, aligning schemas with revised government data-governance requirements and reducing redundant collections by ~30%.
- Engineered the admin module with role-based middleware for content moderation, mentor verification, and workflow approvals, improving operational throughput.
- Consolidated fragmented backend services into a unified Express.js layer with a standardized request/response contract and structured request-tracing logs.

Software Development Intern

Mar 2025 – May 2025

Taurus Hard & Soft Solution — Remote

- Built an EV Vehicle Sales Dashboard backend with Node.js and Express.js, exposing REST endpoints for sales ingestion, aggregation, and reporting.
- Designed a CSV bulk-upload pipeline with streaming parsing, schema validation, and batched MySQL inserts, cutting manual data-entry effort for the sales team.
- Modelled normalized MySQL schemas and wrote optimized aggregation queries powering a React + Chart.js analytics frontend.

PROJECTS

Sarthi — AI-Powered Learning Platform

[GitHub](#)

Node.js, Express.js, MongoDB, FFmpeg, OpenAI Whisper, OpenRouter, Cloudinary, JWT

- Architected an end-to-end **video-to-notes-and-quiz** pipeline: on mentor upload, the backend extracts audio with **FFmpeg**, transcribes via **Whisper**, and forwards timestamped transcript segments to **OpenRouter** LLMs to generate structured notes and quizzes.
- Persisted timestamp-indexed transcript chunks in MongoDB so quizzes are **time-synced to the lecture** — auto-triggered to the student at the exact moment of the corresponding video segment during playback.
- Designed scalable MongoDB schemas with compound indexing and validation for users, courses, transcripts, and quiz attempts to support efficient content-based queries.
- Implemented **JWT authentication** with role-based access control (mentor / student / admin) and refresh-token rotation; integrated **Cloudinary** with signed URLs for optimized media storage and retrieval.
- Decoupled long-running transcription and LLM generation into a background-job flow so upload APIs return immediately, improving UX and isolating failure modes.

Connexus — Alumni & Mentorship Portal

[GitHub](#)

Node.js, Express.js, MongoDB Atlas, REST APIs, Cloudinary, JWT

- Built a backend-first platform with modular Express routing, middleware pipelines, and a clean service/repository separation for scalability and testability.
- Engineered an **agent-request audit trail**: every AI/agent API call is logged with request, response, user, and timestamp into a dedicated history collection, enabling later review of prompt-injection or security breaches.
- Designed MongoDB Atlas schemas with validation rules and compound indexes; applied query-optimization strategies (projection, lean queries, aggregation pipelines) to reduce response time on heavy endpoints.
- Integrated **Cloudinary** with signed uploads, JWT-based auth with RBAC (student/alumni/admin), and request-rate limiting to harden public endpoints.
- Shipped a **Donation Portal**, email-based **Notification System**, and a **Star Alumni** recognition module to drive engagement and contributions.

ACHIEVEMENTS

Smart India Hackathon 2024 — National Winner

Dec 2024

- Led **Team Connexus** to a national win and **₹50,000** cash prize for building Mentor Connect — scheduling, real-time chat, referrals, group discussions, quizzes, admin verification, and government-management tools.

HackHeritage 2024 — Inter-College Finalist

Sept 2024

- Extended Connexus with a Donation Portal, email-driven Notification System, and Star Alumni recognition feature to strengthen alumni engagement.

EDUCATION

Heritage Institute of Technology, Kolkata

Aug 2022 – May 2026

Bachelor of Technology in Information Technology

CGPA: 8.63 / 10

Coursework complete — available to start immediately